

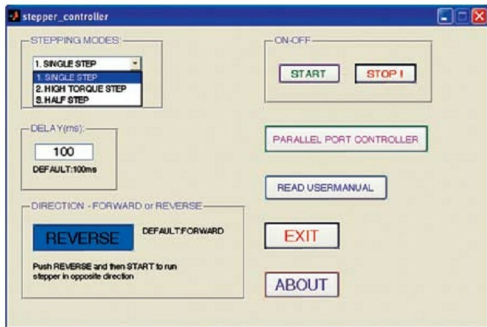


Fig. 7: Screenshot of the program output of stepper motor controller

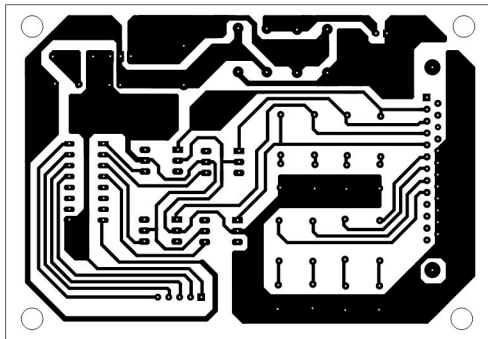


Fig. 8: Actual-size, single-side PCB for the parallel-port controller using Matlab

interface design environment') is similar to Visual Basic (VB) environment. We know that VB is an event-driven programming language which helps users to implement graphic interfaces.

Guide provides a set of tools for creating graphical user interfaces (GUIs). Just type 'Guide' against the prompt in Matlab command window. Select 'Blank GUI' option in the next window to launch Guide.

Using 'Layout' editor option from Guide, you can populate a GUI by

clicking and dragging GUI components—such as axes, panels, buttons, text fields, sliders and so on—into the layout area. You can also create menus and context menus for the GUI. From the layout editor, you can resize the GUI, modify and align components, set tab order, view a hierarchical list of the component objects and set other GUI options.

Guide provides controls like push-button, slider, radio button, checkbox, edit text, static text, pop-up menu, list

box, toggle-button, table, axes, panel, button group and ActiveX control. It even generates a skeleton M-file (plus figure file with a '.fig' extension) which programmers can further use to link other events through 'callback' function. The callback may be an event after mouse click on a button.

'callback' function is used in 'data_port_controller.m' program as given below:

```
function checkbox1_Callback(hObject,  
 eventdata, handles)
```

'checkbox1_Callback()' is a callback function executed while you check and uncheck the checkboxes corresponding to LED1 through LED8 as shown in Fig. 3.

Variables can be used without declaration:

```
handles.output = hObject;  
handles.il=handles.il;  
data  
parport  
guidata(hObject, handles);
```

Here, 'handles.il' is a global variable. Every click on the checkbox makes 'handles.il' toggled. 'data' is a script file which has only one line of code:

```
handles.data = handles.il * 1 +  
handles.i2 * 2 + handles.i3 * 4 +  
handles.i4 * 8 + handles.i5 * 16 +  
handles.i6 * 32 + handles.i7 * 64 +  
handles.i8 * 128 ;
```

'handles.data' is a global variable which updates a new value and then 'parport.m' script file is executed to output the value at the parallel port. Script file executes a sequence of commands rather than accepting or returning parameters. For example, the file generated by Guide is not a script file but a function m-file. Examples of function file include most commands in Matlab, such as sine and plot.

Two more important commands are 'set' and 'get.' For example, 'checkbox1' value sets 'handles.il' as follows:

```
set(handles.checkbox1, 'value',  
 handles.il);
```

In 'stepper_controller.m' file, the direction control variable returns the toggle state of 'togglebutton1' and